

RIZKI SAPUTRA SEMBIRING

Robotics & Automation Engineer | IoT and Industry 4.0 Specialist

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cracktiest.github.io/portfolio/ | Majalengka, West Java

I am a **Robotics & Automation Supervisor** with experience leading multidisciplinary engineering teams and delivering solutions across robotics, AMR, industrial automation, AI computer vision, IoT, embedded systems, and smart manufacturing. As the **Best Graduate in Electrical Engineering** from Universitas Muria Kudus, I have contributed to industrial technology projects at Gistex, Djarum, and Kereta Api Indonesia, focusing on transforming engineering concepts into scalable, production-ready solutions. My experience includes robotic systems, RFID-based production tracking, industrial IoT, automated manufacturing equipment, renewable energy and digital transformation initiatives that improve operational efficiency and reduce implementation costs.

EDUCATION

Universitas Muria Kudus

Bachelor in Electrical Engineering | GPA: 3.72 / 4.0

Kudus
2016-2020

Activities: Robotic

- Completed my research to develop cigarette production machine for automatic control at Djarum Oasis Kretek Factory
- Community Service Program in Tondokerto, Pati
- Field work study at LEN Industri & Badan Tenaga Nuklir Nasional

WORK EXPERIENCE

Gistex

Robotic Automation Engineer Supervisor

Majalengka
Mar 2025 - Present

- Lead and supervise a multidisciplinary engineering team developing robotics, AMR, industrial automation, AI computer vision, IoT, embedded systems, and smart manufacturing solutions.
- Manage end-to-end engineering projects from concept development and system architecture through prototyping, testing, validation, deployment, and production implementation.
- Lead the integration and deployment of robotic arms, AMR systems, automated sewing equipment, pneumatic automation, and custom industrial machinery to improve production efficiency and operational consistency.
- Develop smart manufacturing solutions involving RFID garment tracking, industrial IoT, real-time monitoring, MQTT communication, embedded control, and production data integration.
- Lead technical recruitment, mentoring, engineering capability development, and cross-functional collaboration across R&D, Production, Industrial Engineering, IT, and operational teams.
- Manage project scope, budgets, timelines, engineering resources, technical risks, and stakeholder coordination to ensure successful project delivery.
- Expand engineering initiatives into renewable energy, contributing to the installation and IoT-based monitoring of a 1.2 kWp solar photovoltaic system.
- Drive Industry 4.0 and Industry 5.0 transformation through the integration of intelligent automation, robotics, AI, connected systems, and human-centered manufacturing technologies.

Robotic Automation Engineer

Jan 2022 – Mar 2025

- Develop and deploy computer vision solutions to support company objectives.
- Optimize resource utilization, including hardware and budgets.
- Designed and implemented IoT and automation systems, covering hardware selection, firmware development, and cloud integration.
- Led smart factory and productivity improvement projects, utilizing IoT technologies to enhance operator efficiency and monitor performance.
- Reduced automation and IoT implementation costs by approximately 1 billion IDR compared to market alternatives.
- Built and maintained web and desktop applications, improving UX and functionality.
- Created and optimized databases for enhanced query performance and data integrity.

Kereta Api Indonesia

Signalling, Telecommunication, and Electricity Engineer Intern

Bandung

Mar 2020 – Aug 2020

- Participated in an internship program facilitated by Forum Human Capital Indonesia BUMN.
- Collaborated with the Sintelis core engineering team at Kereta Api Indonesia HQ.
- Gained hands-on experience in electrical-mechanical signalling systems and CBTC (Communication-Based Train Control) technology.
- Learned about the design and operation of railway communication and electrical systems.

Djarum

Research Intern

Kudus

Jan 2020 – Feb 2020

- Research to develop one of the cigarette tip paper processing production machines into auto system as thesis project

Djarum

Electrical Engineer Intern

Kudus

Jul 2018 – Sep 2018

- Collaborated with senior engineers to design, install, and maintain control systems for production machinery and processes.
- Conducted testing and tuning of electrical systems, including control panels, sensors, and automation components.
- Successfully upgraded a control panel and vision sensor system to detect defective cigarette excise bands with costs over 2 billion IDR.
- Gained practical experience in electronic control systems, instrumentation, and automation within one of Indonesia's largest tobacco industries.

Dinas Perhubungan Kabupaten Kudus

IT Support Intern

Kudus

Jul 2014 – Sep 2014

- Supported the management and maintenance of network infrastructure, including routers, switches, firewalls, and wireless access points (WAP).
- Handled the procurement, configuration, and maintenance of computer hardware, software, and peripherals.

NON-FORMAL EDUCATION

Cisco Networking Academy

CISCO

Kudus

2013 – 2015

- Focused on designing, configuring, and troubleshooting network infrastructures for small-to-medium businesses and ISPs.
- Developed expertise in IP addressing, VLANs, routing protocols, and network security for scalable solutions.
- Covered foundational networking concepts, including designing, and implementing secure wired and wireless networks for homes and small businesses.
- Emphasized practical skills in troubleshooting, network optimization, and securing data transmissions.

CERTIFICATION

- FreeRTOS, PT. Nusabot Inovasi Teknologi (2024)
- Program Magang Mahasiswa Bersertifikat, Forum Human Capital Indonesia (2020)
- CCNA Discovery: Working at a Small-to-Medium Business or ISP, CISCO (2015)
- CCNA Discovery: Networking for Home and Small Businesses, CISCO (2014)

AWARDS

- Best Employee (2024): Recognized for contributions to digitalization, innovation, and technology at Gistex
- Best Graduate (2020): Graduated as the top student in Electrical Engineering
- Academic Achievement Scholarship (2018): 1-year academic development scholarship funded by the Ministry of Research, Technology, and Higher Education for the best parallel students
- Highest Test Score of Electrical Engineering in the admission of new students (2016)

SKILL

Soft Skill

- Innovation, Problem-Solving, Leadership, Teamwork, Project Management, Communication.

Technical Skill

- AMR/AGV, Robotic Arm Integration, Industrial Automation, AI Computer Vision, RFID, Industrial IoT, Machine Integration, Renewable Energy, Embedded Systems, Linux Server Configuration, Network Configuration, Instrumentation and Control System, Smart Factory and Digital Transformation.

Software Tools

- Arduino IDE, Visual Studio Code, GitHub, MSSQL, MySQL, C++, C#, Python, PHP, HTML, Node-RED, YOLO, OpenCV, PlatformIO, Linux/Ubuntu, Docker, MikroTik,

LANGUAGES

- English (Professional)
- Indonesian (Native)